

# Unit - 4

(SV)  $A = U \Sigma V^T$   
Singular value decomposition

## Quadratic Form:-

$$Q(x, y) = ax^2 + cy^2 + 2bxy$$

Ex:  $Q(x, y) = 2x^2 + 3y^2 + 4xy$   
 $Q(1, 1) = 9 = \text{scalar}$

## \* Classifications of Q form:-

1) Positive definite - min

$$Q(x) > 0 \text{ for all } x \neq 0$$

2) Positive semi definite:  $Q(x) \geq 0$   
all x

3) Negative definite - max

$$Q(x) < 0 \text{ for all } x \neq 0$$

4) Negative semi definite:  $Q(x) \leq 0$   
all x

5) Indefinite - saddle

$$Q(x) < 0 \text{ or } Q(x) > 0$$

## \* Matrix Notation of QF:-

$$Q(x) = x^T A x$$

A = symmetric matrix

$$Q(x, y) = ax^2 + cy^2 + 2bxy$$

$$= \begin{pmatrix} x & y \end{pmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \begin{pmatrix} x & y \end{pmatrix} \begin{bmatrix} ax+by \\ bx+cy \end{bmatrix}$$

$$= ax^2 + bxy + bxy + cy^2$$

$$Q(x, y) = ax^2 + cy^2 + 2bxy$$

Every sym matrix can be written in quadratic form.

$$A = \begin{pmatrix} 2 & -4 \\ -4 & 6 \end{pmatrix} \Rightarrow Q(x) = x^T A x$$

$$= 2x_1^2 + 6x_2^2 - 8x_1x_2$$

Q. Find the matrix of the quadratic form.

$$x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 - 2x_1x_3$$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

Q.  $x_1^2 + x_2^2 + x_3^2$  - pure quadratic no cross product.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

## \* Positive Definite Matrix:-

If  $Q(x) > 0$  for all  $x \neq 0$ , and  $Q(x) = x^T A x$ , then A is called Positive definite Matrix.

When  $Q(x) = ax^2 + cy^2 + 2bxy$  is +ve definite?

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

Cond<sup>n</sup> are  $a > 0$

$$c > 0$$

$$ac - b^2 > 0$$

Express  $Q(x)$  as sum of squares terms,

$$Q(x) = a \left( x + \frac{by}{a} \right)^2 + \left( \frac{ac - b^2}{a} \right) y^2$$

$$a \left( x + \frac{b}{a} y \right)^2 + \left( c - \frac{b^2}{a} \right) y^2$$

a = pivot

$$\frac{ac - b^2}{a} = \text{pivot}$$

$\Rightarrow$  tests

sum of sq

Pivot

Eigen val

det test

Problem 6.1

Let  $A$  be a  $3 \times 3$  matrix  $A_1$  &  $A_2$  correspond to  $f_1, f_2$

$$f_1 = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$

$$f_2 = x_1^2 + 2x_2^2 + 11x_3^2 - 2x_1x_2 - 2x_1x_3 - 4x_2x_3$$

$$A = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & -2 \\ -1 & -2 & 11 \end{bmatrix}$$

$\begin{bmatrix} a & b \\ b & c \end{bmatrix}$  for +ve def

$$a > 0$$

$$c > 0$$

$$ac - b^2 > 0$$

$$\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \times \begin{bmatrix} -1 & 2 \\ 3 & -8 \end{bmatrix} \times$$

$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$  is +ve def

Let  $A^{-1} = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$  for +ve def

$$\begin{bmatrix} c & -b \\ -b & a \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$$

also def.  $A^{-1} = \frac{1}{10-9} \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$$

is +ve def

6.2

①  $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

Pivot method: convert to U  
Pivots  $> 0$   
 $\therefore$  It is +ve def

$$\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & -3/2 & 3/2 \end{bmatrix}$$

$\therefore$  +ve semi-definite

$$U = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & 0 & 0 \end{bmatrix}$$

②  $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$

Det method  
Det of  $B_i > 0$

$$B_1 = (2)$$

$$B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$B_3 = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}$$

$$|B_1| = 2 > 0$$

$$|B_2| = 3 > 0$$

$$|B_3| = 2(2) + 1(-2+1) - 1(-1+2)$$

$$= 6 + (-1) - 1(1)$$

$$= 4 > 0$$

+ve definite.

③  $C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$

Eigen value method.  
 $D = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$

$$\begin{vmatrix} -\lambda & 1 & 2 \\ 1 & -\lambda & 1 \\ 2 & 1 & -\lambda \end{vmatrix} = 0$$

$$-\lambda(\lambda^2 - 1) - 1(-\lambda - 2) + 2(1 + 2\lambda) = 0$$

$$-\lambda^3 + \lambda + \lambda + 2 + 2 + 4\lambda = 0$$

$$6\lambda + 4 - \lambda^3 = 0$$

$$\lambda = 1 \pm \sqrt{3}, -2$$

$$\lambda = \begin{matrix} 1+\sqrt{3} \\ 1-\sqrt{3} \\ -2 \end{matrix}$$

eigen value of  $c$  are  
square of  $(1+\sqrt{3})^2$ ,  
 $(1-\sqrt{3})^2$ ,  $(-2)^2 > 0$   
 $\therefore c$  is positively definite.

Ex 1. the pivots are  
2,  $3/2$ ,  $4/3$   
Sum of squares.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

1<sup>st</sup> test: pivot method.

$$\begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow{R_3 + \frac{2}{3}R_2} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & 0 & 4/3 \end{bmatrix}$$

$\Rightarrow$  positive def.

2<sup>nd</sup> test:  $|2| = 2 > 0$

$$\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3 > 0$$

$$\begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} = 4 > 0$$

$$2(3) + (-2) + 0 = 6 - 2 = 4$$

$\Rightarrow$  positive def.

3<sup>rd</sup> test:  $\begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} = 0$

$$(2-\lambda)[(2-\lambda)^2 - 1] + 1(-2+\lambda) + 0$$

$$(2-\lambda)^3 - 2 + \lambda - 2 + \lambda = 0$$

$$(2-\lambda)[(2-\lambda)^2 - 1] - 1(2-\lambda) = 0$$

$$(2-\lambda)[(2-\lambda)^2 - 1 - 1] = 0$$

$$(2-\lambda)[4 + \lambda^2 - 4\lambda - 1 - 1] = 0$$

$$\lambda^2 + 2\lambda^2 - 6\lambda - 2 - 2 - 4\lambda - \lambda^3 + 4\lambda^2 + \lambda + \lambda = 0$$

$$-\lambda^3 + 6\lambda^2 - 8\lambda + 4 = 0$$

$$\lambda_1 = 4.38, 0.808 \pm 0.508i,$$

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

as sum of squares

$$x^T A x = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3$$

$$A = LDL^T \quad (\because A \text{ symm})$$

$$x^T A x = (x^T L) D (L^T x) \rightarrow \textcircled{1}$$

$$A \rightarrow U$$

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow{R_2 + \frac{1}{2}R_1} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_3 + \frac{2}{3}R_2}$$

$$U = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & 0 & 4/3 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & -2/3 & 1 \end{bmatrix}$$

$$\text{Now, } L^T x = \begin{pmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -2/3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$L^T x = \begin{pmatrix} x_1 - x_2/2 \\ x_2 - 2x_3/3 \\ x_3 \end{pmatrix}$$

$$(L^T x)^T = x^T L = \begin{pmatrix} x_1 - x_2/2 & x_2 - 2x_3/3 & x_3 \end{pmatrix}$$

from eq 1,

$$x^T A x = \begin{pmatrix} x_1 - x_2/2 & x_2 - 2x_3/3 & x_3 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 4/3 \end{pmatrix}$$

$$\begin{pmatrix} x_1 - x_2/2 \\ x_2 - 2x_3/3 \\ x_3 \end{pmatrix}$$

$$x^T A x = \begin{pmatrix} x_1 - x_2/2 & x_2 - 2x_3/3 & x_3 \end{pmatrix} \begin{pmatrix} 2(x_1 - x_2/2) \\ 3/2(x_2 - 2x_3/3) \\ 4/3(x_3) \end{pmatrix}$$

$$2(x_1 - \frac{x_2}{2})^2 + \frac{3}{2}(x_2 - \frac{2x_1}{3})^2 + \frac{4x_2^2}{3}$$

Result of least squares.

$A^T A$  - Sym of +ve def.

$$A x = b$$

$$A^T A \hat{x} = A^T b$$

$A = R^T R$  → if it's possible to express like this then  $A$  is positive def.

$R$  has indep col  
elimination:  $A = L D L^T$   
 $= (L \sqrt{D}) (\sqrt{D} L^T)$   
 $= R^T R$

$$\Rightarrow \boxed{R = \sqrt{D} L^T}$$

$$A^+ = S \Lambda^+ S^T$$

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \quad \sqrt{D} = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{3} \end{pmatrix}$$

$$D^T = D$$

Eigen value

$$A = Q \Lambda Q^T$$

$$= (Q \sqrt{\Lambda}) (\sqrt{\Lambda} Q^T)$$

$$= R^T R$$

$$\boxed{R = \sqrt{\Lambda} Q^T}$$

Cholesky's decomposition

The decomposition of  $A$ .  $A = R^T R$

same matrix  $\Sigma$  its transpose  $A^T$

$$3) R = Q \sqrt{\Lambda} Q^T$$

$$4) A = Q R$$

$$R = Q^T A$$

$$Q = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

1st test: pivot

$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \xrightarrow[R_3 + R_2]{R_1 + R_2} \begin{pmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & -3/2 & 3/2 \end{pmatrix} \xrightarrow{R_1 + R_2} \begin{pmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & 0 & 0 \end{pmatrix}$$

∴ +ve semi definite

2nd test: Det

$$|2| = 2 > 0$$

$$\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3 > 0$$

$$\begin{vmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{vmatrix} = 2(1) + 1(2-1) - 1(1+2) = 2 + 1 - 3 = 0 = 0$$

∴ +ve semi definite

3rd test: eigen

$$\begin{vmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & -1 \\ -1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)[(2-\lambda)^2 - 1] + 1[-1(2-\lambda) - 1] - 1[1 + (2-\lambda)] = 0$$

$$(2-\lambda)^3 - (2-\lambda) - 1 - 1 + (2-\lambda) = 0$$

$$(2-\lambda)^3 - (2-\lambda) - 2 = 0$$

$$8 - 6\lambda^2 + 12\lambda - 2 + \lambda - 2 = 0$$

$$-2\lambda^3 + 6\lambda^2 - 11\lambda + 4 = 0$$

$$\lambda = 3, 0$$

Semi +ve definite

Q. From the pivots, eigen values, and eigen vectors of  $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$

write  $A$  as  $R^T R$  in three ways,

$$(L\sqrt{D})(\sqrt{D}L^T)$$

$$(Q\sqrt{\Lambda})(\sqrt{\Lambda}Q^T)$$

$$(Q\sqrt{\Lambda}Q^T)(Q\sqrt{\Lambda}Q^T)$$

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \xrightarrow{R_2 - \frac{4}{5}R_1} \begin{bmatrix} 5 & 4 \\ 0 & 9/5 \end{bmatrix} = U$$

$$\begin{array}{l} 5 - \frac{4}{5}(4) \\ 5 - \frac{16}{5} \\ \frac{25-16}{5} \end{array} \quad \downarrow \quad \begin{array}{l} R_1 - \frac{4}{9}(R_2) \\ 5 - \frac{4}{9}(9/5) \\ 5 - \frac{4}{5} \end{array} \quad \begin{bmatrix} 5 & 0 \\ 0 & 9/5 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 \\ \frac{4}{5} & 1 \end{bmatrix}$$

$$[A - \lambda I] = 0$$

$$\begin{vmatrix} 5-\lambda & 4 \\ 4 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)^2 - 16 = 0$$

$$25 + \lambda^2 - 10\lambda - 16 = 0$$

$$\lambda^2 - 10\lambda + 9 = 0$$

$$\lambda(\lambda-9) - 1(\lambda-9)$$

$$\lambda = 9 \quad \lambda = 1$$

$$\downarrow \quad \downarrow$$

$$x_1 = \quad x_2 =$$

$$\lambda_1 = 9$$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 5-9 & 4 \\ 4 & 5-9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\begin{pmatrix} -4 & 4 \\ 4 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$-4x_1 + 4x_2 = 0$$

$$4x_1 - 4x_2 = 0$$

$$x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$4x_1 + 4x_2 = 0$$

$$x_1 = -x_2$$

$$x_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\Rightarrow (L\sqrt{D})(\sqrt{D}L^T)$$

$$L = \begin{pmatrix} 1 & 0 \\ 4/5 & 1 \end{pmatrix}$$

$$R = \sqrt{D}L^T$$

$$R = \sqrt{D}L^T = \begin{bmatrix} \sqrt{5} & 0 \\ 0 & 3/\sqrt{5} \end{bmatrix} \begin{bmatrix} 1 & 4/5 \\ 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} \sqrt{5} & 4/\sqrt{5} \\ 0 & 3/\sqrt{5} \end{bmatrix}$$

$$\Rightarrow (Q\sqrt{\Lambda})(\sqrt{\Lambda}Q^T)$$

$$R = \sqrt{\Lambda}Q^T$$

$$Q = \begin{bmatrix} q_1 & q_2 \end{bmatrix}, \quad q_1 = \frac{x_1}{\|x_1\|}$$

$$q_2 = \frac{x_2}{\|x_2\|}$$

$$Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \quad \sqrt{\Lambda} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$R = \begin{bmatrix} 3/\sqrt{2} & 3/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$\Rightarrow (Q\sqrt{\Lambda}Q^T) = R$$

$$\sqrt{\Lambda}Q^T = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

$$Q\sqrt{\Lambda}Q^T = \begin{pmatrix} 3/\sqrt{2} & 3/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{3}{2} - \frac{1}{2} & \frac{3}{2} - \frac{1}{2} \\ \frac{3}{2} - \frac{1}{2} & \frac{3}{2} - \frac{1}{2} \end{pmatrix}$$

$$R = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 3/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 3/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$$

SVD  
Singular value decomposition

$$A = S \Lambda S^T$$

$$Q \Lambda Q^T$$

$$A = Q \Lambda Q^T$$

$$A = U \Sigma V^T \rightarrow \text{SVD}$$

orthogonal  
diagonal  
orthogonal

A can be any square or rect

$$U \text{ is } A A^T$$

$$V \text{ is } A^T A$$

Eigen value same if sq if rect then the eigen values are same rest 0.

$\Sigma$  = diagonal matrix same dimension of A.

arrange singular values of matrix A.

Eigen value in  $\Sigma$ .

$$\lambda_1, \lambda_2, \lambda_3, \dots$$

$$\sqrt{\lambda_1}, \sqrt{\lambda_2}, \sqrt{\lambda_3}, \dots$$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots \\ 0 & \sigma_2 & \dots \\ 0 & 0 & \sigma_3 \dots \end{bmatrix}$$

Symm

Eigen value - diff

Eigen vectors - Orthogonal - mat

Q. find the SVD of matrix

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix}$$

$$A A^T = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

Eigen value,

$$\begin{vmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{vmatrix} = 0$$

$$(17-\lambda)^2 - 64 = 0$$

$$17^2 + \lambda^2 - 34\lambda - 64 = 0$$

$$\lambda^2 - 34\lambda + 225 = 0$$

$$\lambda = 25, \lambda = 9$$

$$\lambda_1 = 25$$

$$\begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-8x_1 + 8x_2 = 0$$

$$8x_1 - 8x_2 = 0$$

$$x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 9$$

$$\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$8x_1 + 8x_2 = 0$$

$$x_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$x_1 \perp x_2$$

$u_1$  = orthonormal vector of  $x_1$

$$u_1 = \frac{x_1}{\|x_1\|}$$

$$u_2 = \frac{x_2}{\|x_2\|}$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$U = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix}$$

$\sigma_1 = \sqrt{\lambda_1}$   $\sigma_2 = \sqrt{\lambda_2}$   
(singular values of matrix A)

$$\Sigma_1 = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix}$$

$$V = A^T A$$

$$A^T A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \\ 2 & -2 & 8 \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \\ 2 & -2 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}$$

$$(A^T A - \lambda I)(*) = 0$$

$$\begin{vmatrix} 10-\lambda & 12 & 2 \\ 12 & 13-\lambda & -2 \\ 2 & -2 & 8-\lambda \end{vmatrix} = 0$$

$$(10-\lambda)[(13-\lambda)(8-\lambda) - 4] - 12[-12\lambda + 4] + 2[-24 - 2(13-\lambda)] = 0$$

$$(10-\lambda)(\lambda^2 - 21\lambda + 100 - 4) - 12[-12\lambda + 4] + 2[-24 - 26 + 2\lambda] = 0$$

$$\lambda = 25, 9, 0$$

$$\lambda_1 = 25$$

$$\begin{bmatrix} -15 & 12 & 2 \\ 12 & -12 & -2 \\ 2 & -2 & -17 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x}{-15} = \frac{y}{12} = \frac{z}{2}$$

$$\begin{vmatrix} -12 & -2 \\ -2 & -17 \end{vmatrix} \quad \begin{vmatrix} -12 & 12 \\ -17 & 2 \end{vmatrix} \quad \begin{vmatrix} 12 & -12 \\ 2 & -2 \end{vmatrix}$$

$$\frac{x}{200} = \frac{y}{200} = \frac{z}{-24+24} = 0 \quad \|x_3\|$$

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{0}$$

$$V_1 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \quad x_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} 4 & 12 & 2 \\ 12 & 4 & -2 \\ 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x}{4} = \frac{y}{4} = \frac{z}{2}$$

$$\begin{vmatrix} 4 & -2 \\ -2 & 1 \end{vmatrix} \quad \begin{vmatrix} -2 & 12 \\ -1 & 2 \end{vmatrix} \quad \begin{vmatrix} 12 & 4 \\ 2 & -2 \end{vmatrix}$$

$$\frac{x}{-8} = \frac{y}{8} = \frac{z}{-3+2}$$

$$\frac{x}{-1} = \frac{y}{1} = \frac{z}{-4}$$

$$x_2 = \begin{pmatrix} -1 \\ 1 \\ -4 \end{pmatrix}$$

$$V_2 = \frac{x_2}{\|x_2\|} = \frac{x_2}{\sqrt{18}} = \frac{x_2}{3\sqrt{2}}$$

Eg  $x_1$   $x_2$  not orthogonal  
make it orthogonal  
and then normalize  
By Gramschist

$$V_2 = \begin{pmatrix} -1/3\sqrt{2} \\ 1/3\sqrt{2} \\ -4/3\sqrt{2} \end{pmatrix}$$

$$\lambda_3 = 0$$

$$\begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x}{13} = \frac{y}{12} = \frac{z}{2}$$

$$\begin{vmatrix} 13 & -2 \\ -2 & 8 \end{vmatrix} \quad \begin{vmatrix} -2 & 12 \\ 8 & 2 \end{vmatrix} \quad \begin{vmatrix} 12 & 13 \\ 2 & -2 \end{vmatrix}$$

$$\frac{x}{100} = \frac{y}{-100} = \frac{z}{50}$$

$$\frac{x}{2} = \frac{y}{-2} = \frac{z}{1}$$

$$x_3 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \quad V_3 = \frac{x_3}{\|x_3\|}$$

$$V_3 = \begin{pmatrix} 2/\sqrt{5} \\ -2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix}$$

$$V = \begin{bmatrix} v_1 & v_2 & v_3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$U^T \Sigma V^T = A$$

$$\Sigma V^T = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1/3\sqrt{2} & 2/3 \\ 1 & 1/3\sqrt{2} & -2/3 \\ 0 & -4/3\sqrt{2} & 1/3 \end{bmatrix}$$

$$\Sigma V^T = \begin{bmatrix} 5 & -5/3\sqrt{2} & 10/3 \\ 3 & 1/\sqrt{2} & -2 \end{bmatrix}$$

$$U^T \Sigma V^T = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 & -5/3\sqrt{2} & 10/3 \\ 3 & 1/\sqrt{2} & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 5/\sqrt{2} - 3/\sqrt{2} & -5/3 - 1/2 & 10/3\sqrt{2} + 2/\sqrt{2} \\ 5/\sqrt{2} + 3/\sqrt{2} & -5/3 + 1/2 & 10/3\sqrt{2} - 2/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 2/\sqrt{2} & -10-3 & 10+6 \\ 8/\sqrt{2} & -10+3 & 10-6 \end{bmatrix}$$

$$= \sqrt{2} \begin{bmatrix} 1 & -13 & 16 \\ 4 & -7 & 4 \end{bmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/3\sqrt{2} & 2/3 \\ 1/\sqrt{2} & 1/3\sqrt{2} & -2/3 \\ 0 & -4/3\sqrt{2} & 1/3 \end{bmatrix}$$

$$U^T = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/3\sqrt{2} & 1/3\sqrt{2} & -4/3\sqrt{2} \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}$$

$$\Sigma V^T = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/3\sqrt{2} & 1/3\sqrt{2} & -4/3\sqrt{2} \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$\Sigma V^T = \begin{bmatrix} 5/\sqrt{2} & 5/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & -4/\sqrt{2} \\ 2/\sqrt{2} & -2/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\Sigma V^T = \begin{bmatrix} 5/\sqrt{2} & 5/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & -4/\sqrt{2} \end{bmatrix}$$

$$U^T \Sigma V^T = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5/\sqrt{2} & 5/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & -4/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 5/2 + 1/2 & 5/2 + 1/2 & 0 \\ -5/2 + 1/2 & -5/2 + 1/2 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 2 & 2 \\ -2 & 2 & -2 \end{bmatrix}$$

Summary:

Find  $AA^T \rightarrow U$   
Find  $V \rightarrow \Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$   
 $\sigma_1 = \sqrt{\lambda_1}$

Find  $A^T A \rightarrow V$   
 $\sigma_1 = \sqrt{\lambda_1}$

$$U^T \Sigma V^T = A$$